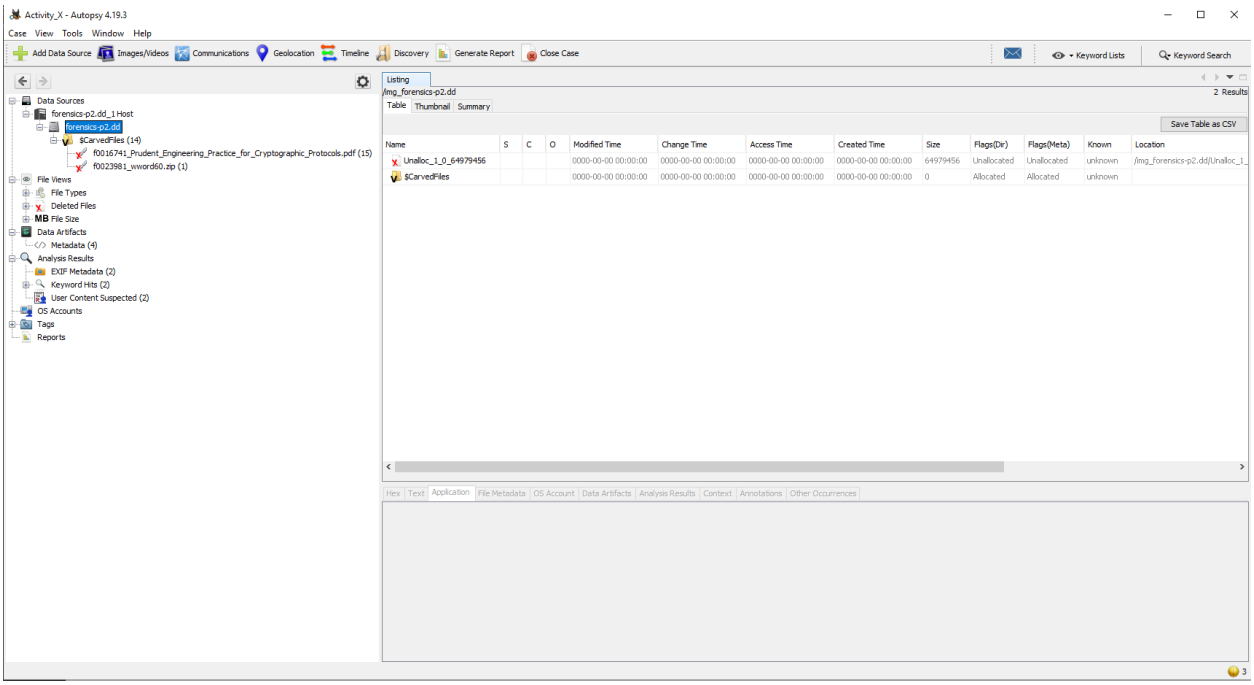
Activity: Computer Forensics

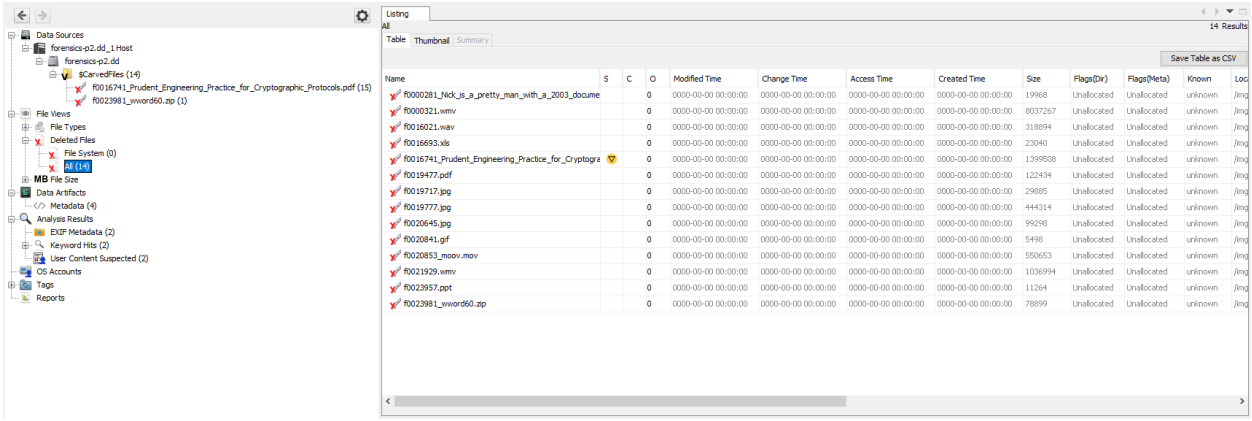
Part I. File Carving

1. Look at the data on the file system (Click on Data Sources and look at the hex values on the right). The file system has no files, but why are we able to find items on the disk image? Explain why the file system has no files but there are items that can be found on the disk image.



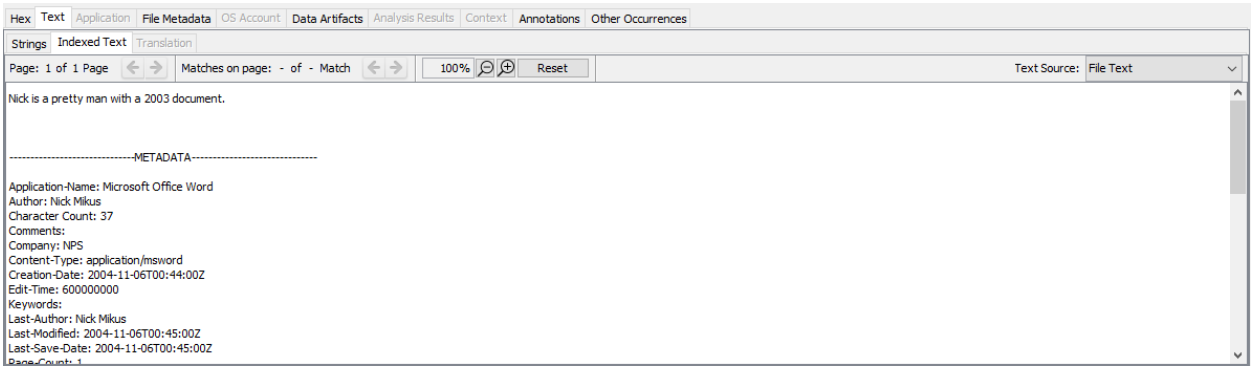
- The files or folders have been marked as unallocated space rather than removed by overwrite those space to “0” or null value when a user commanded to delete them. Still, the files’ content would have remained which could be found without pointer indicated, unless they have been overwritten yet.

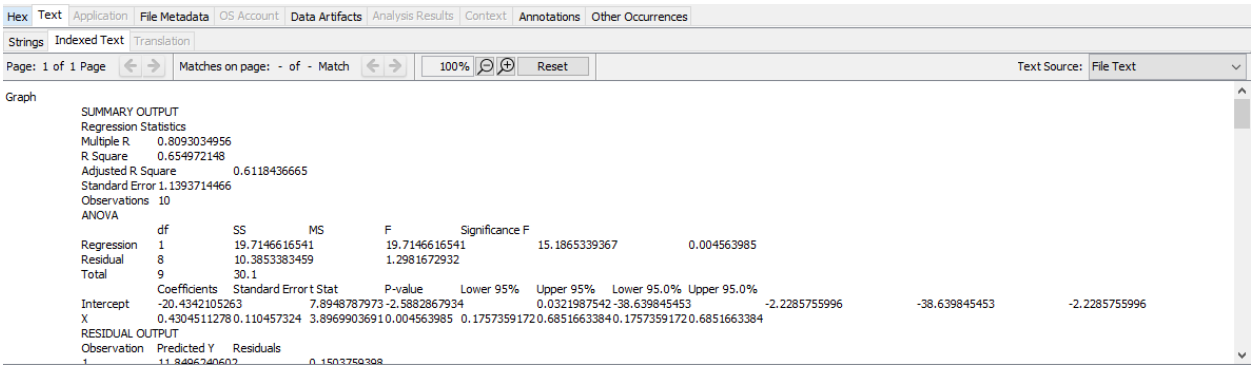
2. How many objects can you find?

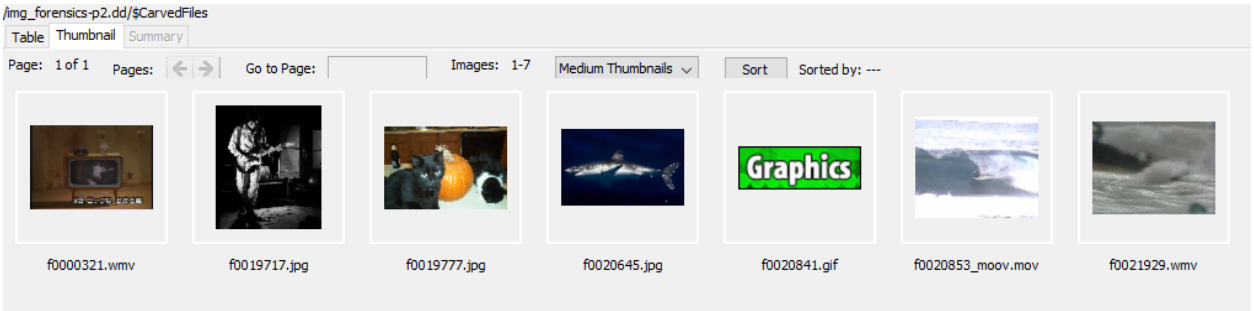


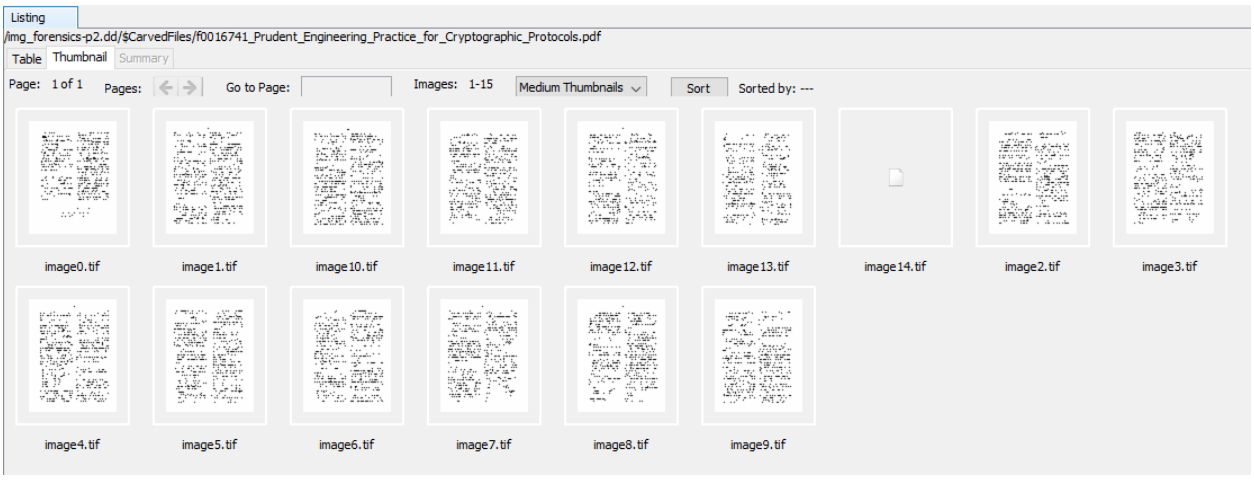
- As said in the document, there were totally 14 objects found deleted from the data source.

3. List all the objects here and report on whether or not the content is accessible or damaged/corrupted. Also note which files were actually already deleted.









- After access attempts, all 14 files were accessible, even though some was partly corrupted. The files were just unallocated their space, but not removed permanently.

4. Think securely: If we want to delete files on a magnetic hard disk and not have them be recovered by any tool, what do we need to do? And how much time do you think you need to wipe a 1TB magnetic hard disk?

- Easily done by overwriting 7 - 35 times on the space where those files were allocated with random bytes.

- For 1TB magnetic hard disk with 100 MB/s writing speed, the files should be completely erased after randomly overwritten 35 times, as the US Government's official data erasure document issued. The 1TB hard disk should take (35 times * 1TB * 1024GB/1TB * 1024MB/1GB / 100 MB/s) = 102 hours approximately.

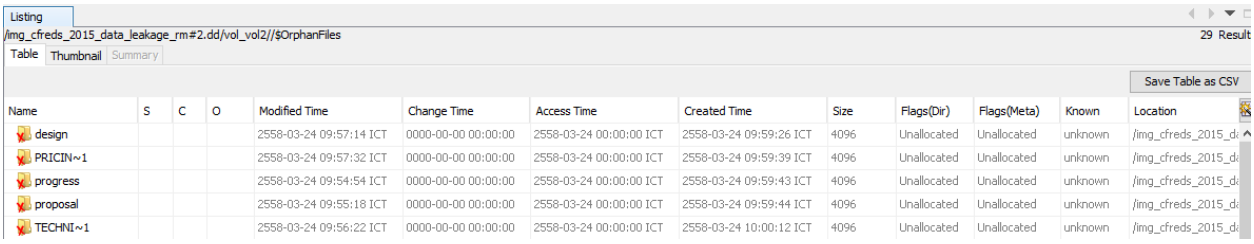
5. Will file carving be able to recover deleted files on an SSD? Why or why not?

- Despite of Trim command utilized on SSD, there is a significance process, named 'Garbage Collection', which keeps on deleting a block with invalid data whenever a

computer is idle. To clarify, the deleted files can be recovered if the Garbage Collection hasn't been activated yet.

Part II. Investigation

1. List all directories that were traversed in 'RM#2'.

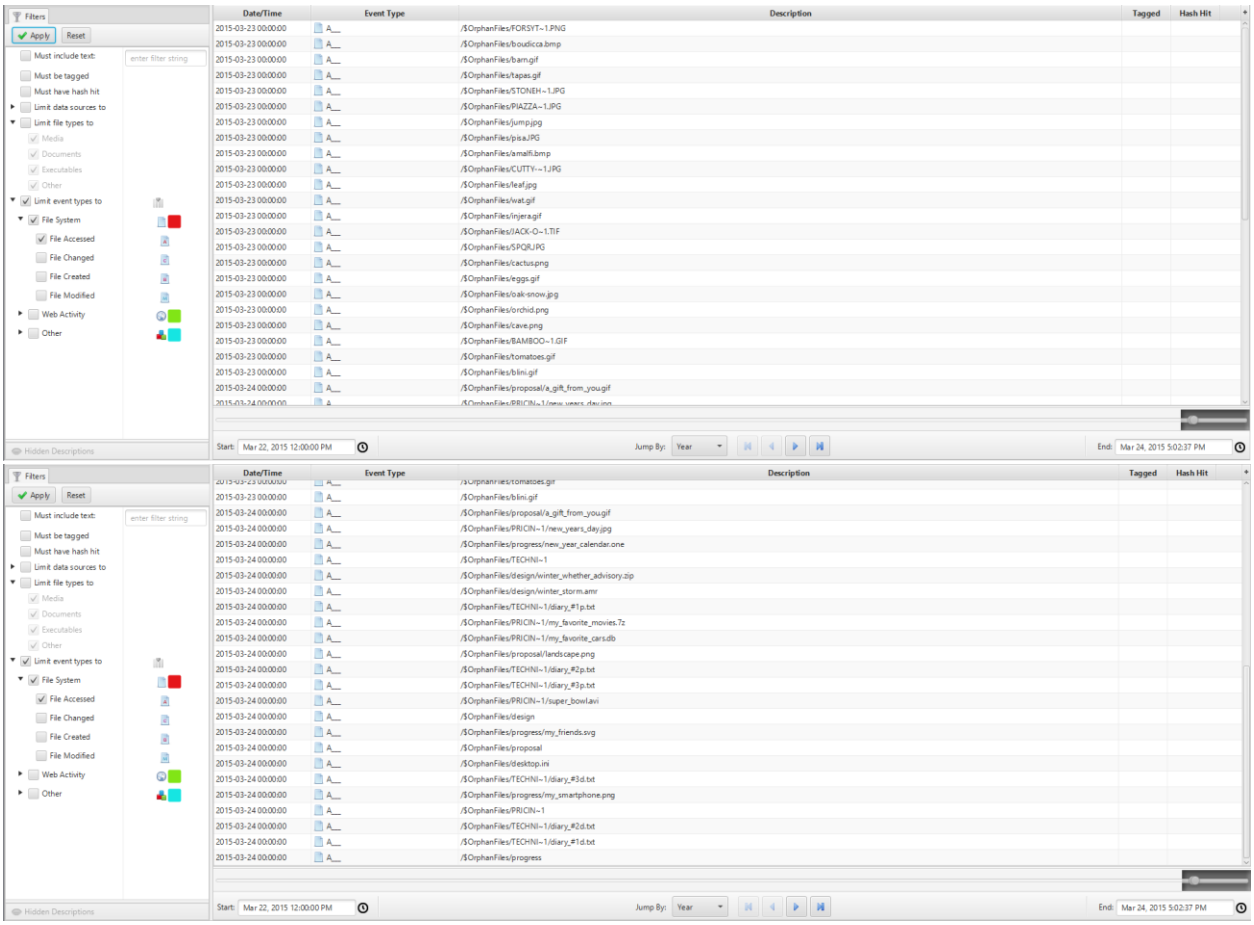


Listing
/img_cfreds_2015_data_leakage_rm#2.dd/vol_vol2//$OrphanFiles — 29 Result
Table | Thumbnail | Summary
Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
design				2558-03-24 09:57:14 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:26 ICT	4096	Unallocated	Unallocated	unknown	/img_cfreds_2015_d
PRICIN~1				2558-03-24 09:57:32 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:39 ICT	4096	Unallocated	Unallocated	unknown	/img_cfreds_2015_d
progress				2558-03-24 09:54:54 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:43 ICT	4096	Unallocated	Unallocated	unknown	/img_cfreds_2015_d
proposal				2558-03-24 09:55:18 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:44 ICT	4096	Unallocated	Unallocated	unknown	/img_cfreds_2015_d
TECHNI~1				2558-03-24 09:56:22 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 10:00:12 ICT	4096	Unallocated	Unallocated	unknown	/img_cfreds_2015_d

- All the traversed ones were thoroughly contained in a volume 2.

2. List all files that were opened in 'RM#2'.



Filters — Apply — Reset
Must include text: enter filter string
Must be tagged
Must have hash hit
Limit data sources to
Limit file types to: Media, Documents, Executables, Other
Limit event types to: File System (File Accessed, File Changed, File Created, File Modified), Web Activity, Other
Hidden Descriptions

Date/Time	Event Type	Description	Tagged	Hash Hit
2015-03-23 00:00:00	A_	/$OrphanFiles/FORSYT~1.PNG		
2015-03-23 00:00:00	A_	/$OrphanFiles/boudicca.bmp		
2015-03-23 00:00:00	A_	/$OrphanFiles/barn.gif		
2015-03-23 00:00:00	A_	/$OrphanFiles/tapas.gif		
2015-03-23 00:00:00	A_	/$OrphanFiles/STONEH~1.JPG		
2015-03-23 00:00:00	A_	/$OrphanFiles/PIAZZA~1.JPG		
2015-03-23 00:00:00	A_	/$OrphanFiles/jump.jpg		
2015-03-23 00:00:00	A_	/$OrphanFiles/pisa.JPG		
2015-03-23 00:00:00	A_	/$OrphanFiles/amalfi.bmp		
2015-03-23 00:00:00	A_	/$OrphanFiles/CUTTY~~1.JPG		
2015-03-23 00:00:00	A_	/$OrphanFiles/leaf.jpg		
2015-03-23 00:00:00	A_	/$OrphanFiles/wat.gif		
2015-03-23 00:00:00	A_	/$OrphanFiles/injera.gif		
2015-03-23 00:00:00	A_	/$OrphanFiles/JACK-O~1.TIF		
2015-03-23 00:00:00	A_	/$OrphanFiles/$PQRJ.PG		
2015-03-23 00:00:00	A_	/$OrphanFiles/cactus.png		
2015-03-23 00:00:00	A_	/$OrphanFiles/eggs.gif		
2015-03-23 00:00:00	A_	/$OrphanFiles/oak-snow.jpg		
2015-03-23 00:00:00	A_	/$OrphanFiles/orchid.png		
2015-03-23 00:00:00	A_	/$OrphanFiles/cave.png		
2015-03-23 00:00:00	A_	/$OrphanFiles/BAMBOO~1.GIF		
2015-03-23 00:00:00	A_	/$OrphanFiles/tomatoes.gif		
2015-03-23 00:00:00	A_	/$OrphanFiles/blini.gif		
2015-03-24 00:00:00	A_	/$OrphanFiles/proposal/a_gift_from_you.gif		
2015-03-24 00:00:00	A_	/$OrphanFiles/PRICIN~1/new_years_day.jpg		

Start: Mar 22, 2015 12:00:00 PM — Jump By: Year — End: Mar 24, 2015 5:02:37 PM

Filters — Apply — Reset
Must include text: enter filter string
Must be tagged
Must have hash hit
Limit data sources to
Limit file types to: Media, Documents, Executables, Other
Limit event types to: File System (File Accessed, File Changed, File Created, File Modified), Web Activity, Other
Hidden Descriptions

Date/Time	Event Type	Description	Tagged	Hash Hit
2015-03-23 00:00:00	A_	/$OrphanFiles/tomatoes.gif		
2015-03-23 00:00:00	A_	/$OrphanFiles/blini.gif		
2015-03-24 00:00:00	A_	/$OrphanFiles/proposal/a_gift_from_you.gif		
2015-03-24 00:00:00	A_	/$OrphanFiles/PRICIN~1/new_years_day.jpg		
2015-03-24 00:00:00	A_	/$OrphanFiles/progress/new_year_calendar.one		
2015-03-24 00:00:00	A_	/$OrphanFiles/TECHNI~1		
2015-03-24 00:00:00	A_	/$OrphanFiles/design/winter_whether_advisory.zip		
2015-03-24 00:00:00	A_	/$OrphanFiles/design/winter_storm.amr		
2015-03-24 00:00:00	A_	/$OrphanFiles/TECHNI~1/diary_#1p.txt		
2015-03-24 00:00:00	A_	/$OrphanFiles/PRICIN~1/my_favorite_movies.7z		
2015-03-24 00:00:00	A_	/$OrphanFiles/PRICIN~1/my_favorite_cars.db		
2015-03-24 00:00:00	A_	/$OrphanFiles/proposal/landscape.png		
2015-03-24 00:00:00	A_	/$OrphanFiles/TECHNI~1/diary_#2p.txt		
2015-03-24 00:00:00	A_	/$OrphanFiles/TECHNI~1/diary_#3p.txt		
2015-03-24 00:00:00	A_	/$OrphanFiles/PRICIN~1/super_bowl.avi		
2015-03-24 00:00:00	A_	/$OrphanFiles/design		
2015-03-24 00:00:00	A_	/$OrphanFiles/progress/my_friends.svg		
2015-03-24 00:00:00	A_	/$OrphanFiles/proposal		
2015-03-24 00:00:00	A_	/$OrphanFiles/desktop.ini		
2015-03-24 00:00:00	A_	/$OrphanFiles/TECHNI~1/diary_#3d.txt		
2015-03-24 00:00:00	A_	/$OrphanFiles/progress/my_smartphone.png		
2015-03-24 00:00:00	A_	/$OrphanFiles/PRICIN~1		
2015-03-24 00:00:00	A_	/$OrphanFiles/TECHNI~1/diary_#2d.txt		
2015-03-24 00:00:00	A_	/$OrphanFiles/TECHNI~1/diary_#1d.txt		
2015-03-24 00:00:00	A_	/$OrphanFiles/progress		

Start: Mar 22, 2015 12:00:00 PM — Jump By: Year — End: Mar 24, 2015 5:02:37 PM

- Display list of files' events filtered by Access Time.

3. Recover deleted files from USB drive 'RM#2'. What files were you able to recover?

60-design
67-PRICIN~1
77-progress
86-proposal
93-TECHNI~1
63-winter_storm
65-winter_whether_advisory
70-my_favorite_cars
72-my_favorite_movies
74-new_years_day
76-super_bowl
80-my_friends
82-my_smartphone
84-new_year_calendar
89-a_gift_from_you
91-landscape
96-diary_#1d
98-diary_#1p
100-diary_#2d
102-diary_#2p
104-diary_#3d
106-diary_#3p
108-desktop
110-amalfi
112-BAMBOO~1
114-barn
116-blini
118-boudicca
120-cactus
122-cave
124-CUTTY-~1
126-eggs
128-FORSYT~1
130-injera
132-JACK-O~1
134-jump
136-leaf
138-oak-snow
140-orchid
142-PIAZZA~1
144-pisa
146-SPQR
148-STONEH~1
150-tapas
152-tomatoes
154-wat
584-f0000000.fat
585-f0000008.fat
586-f0000016_secret_project_revised_points
587-f0028432
588-f0060432.fat
589-f0060440_secret_project_price_analysis_2
590-f0062904
591-f0063104
592-f0083104_secret_project_market_shares
593-f0103200.fat
594-f0103208_secret_project_progress_3
595-f0103328
596-f0112004
597-f0112008
598-f0112064.fat
599-f0112072
600-f0180880
601-f0193552.fat
602-f0193560
603-f0193800
604-f0194696
605-f0197336
606-f0198240_secret_project_technical_review_3
607-f0202856_secret_project_technical_review_3
608-f0203496.fat
609-f0203504.Desktop
610-f0215936
611-f0239392
612-f0259136
613-f0276768
614-f0279296
615-f0284128
616-f0292512
617-f0298560_skip
618-f0317664_skip
619-f0318848
620-f0328544
621-f0330304
622-f0360032.fat
623-f0360064
624-f0361888
625-f0375040
626-f0381600
627-f0385760
628-f0402976
629-f0415040
630-f0431040
631-f0434240
632-f0438720
633-f0454560
634-f0454656
635-f0469408
636-f0473376
637-f0474944
638-f0477632
639-f0494176
640-f0496672
641-f0498336
642-f0500096
643-f0502528
644-f0506912
645-f0511296

- After extracted, all deleted files were recovered successfully. But some files might be corrupted and not viewable.

4. What actions were performed for anti-forensics on USB drive ‘RM#2’?
[Hint: this can be inferred from the results of the above question].

diary_#1d.txt				2558-01-05 17:01:08 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 10:00:12 ICT	121441	Unallocated	Unallocated	unl
diary_#1p.txt				2558-01-05 15:15:08 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 10:00:12 ICT	458267	Unallocated	Unallocated	unl
diary_#2d.txt				2558-01-12 17:25:40 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 10:00:13 ICT	658922	Unallocated	Unallocated	unl
diary_#2p.txt				2558-01-12 15:20:26 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 10:00:14 ICT	1154560	Unallocated	Unallocated	unl
diary_#3d.txt				2558-01-20 16:05:00 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 10:00:15 ICT	2360832	Unallocated	Unallocated	unl
diary_#3p.txt				2558-01-20 14:18:06 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 10:00:18 ICT	325120	Unallocated	Unallocated	unl

Hex | Text | Application | File Metadata | OS Account | Data Artifacts | Analysis Results | Context | Annotations | Other Occurrences

Result: 1 of 1 Result Metadata

Type	Value	Source(s)
Date Modified	2558-01-20 18:18:05 ICT	org.sleuthkit.autopsy.keyword
Program Name	Microsoft PowerPoint	org.sleuthkit.autopsy.keyword
Date Created	2544-12-12 16:10:54 ICT	org.sleuthkit.autopsy.keyword
User ID	company	org.sleuthkit.autopsy.keyword
Owner	company	org.sleuthkit.autopsy.keyword
Source File Path	/img_cfreds_2015_data_leakage_rm#2.dd/vol_vol2/$OrphanFiles/TECHNI~1/diary_#3p.txt	
Artifact ID	-9223372036854775765	

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Kn
winter_whether_advisory.zip				2557-12-16 12:10:26 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:37 ICT	16381123	Unallocated	Unallocated	unl
winter_storm.amr				2558-01-23 16:47:10 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:27 ICT	14547968	Unallocated	Unallocated	unl
wat.gif				2556-05-07 15:12:28 ICT	0000-00-00 00:00:00	2558-03-23 00:00:00 ICT	2558-03-23 16:55:37 ICT	32186	Unallocated	Unallocated	unl
tomatoes.gif				2556-05-07 15:10:50 ICT	0000-00-00 00:00:00	2558-03-23 00:00:00 ICT	2558-03-23 16:55:37 ICT	2240548	Unallocated	Unallocated	unl
tapas.gif				2556-05-07 15:11:10 ICT	0000-00-00 00:00:00	2558-03-23 00:00:00 ICT	2558-03-23 16:55:36 ICT	2242264	Unallocated	Unallocated	unl
super_bowl.avi				2557-12-02 13:28:58 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:40 ICT	10289152	Unallocated	Unallocated	unl
proposal				2558-03-24 09:55:18 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:44 ICT	4096	Unallocated	Unallocated	unl
progress				2558-03-24 09:54:54 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:43 ICT	4096	Unallocated	Unallocated	unl
pisa.JPG				2548-04-10 09:49:04 ICT	0000-00-00 00:00:00	2558-03-23 00:00:00 ICT	2558-03-23 16:55:35 ICT	847709	Unallocated	Unallocated	unl
orchid.png				2556-01-22 14:30:24 ICT	0000-00-00 00:00:00	2558-03-23 00:00:00 ICT	2558-03-23 16:55:33 ICT	8455527	Unallocated	Unallocated	unl
oak-snow.jpg				2553-01-24 15:45:00 ICT	0000-00-00 00:00:00	2558-03-23 00:00:00 ICT	2558-03-23 16:55:32 ICT	1370140	Unallocated	Unallocated	unl
new_years_day.jpg				2557-12-01 14:50:26 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:39 ICT	10237535	Unallocated	Unallocated	unl
new_year_calendar.one				2558-01-12 14:23:42 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:44 ICT	27414	Unallocated	Unallocated	unl
my_smartphone.png				2558-01-05 11:57:22 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:43 ICT	4440235	Unallocated	Unallocated	unl
my_friends.svg				2558-01-20 11:13:44 ICT	0000-00-00 00:00:00	2558-03-24 00:00:00 ICT	2558-03-24 09:59:43 ICT	58368	Unallocated	Unallocated	unl

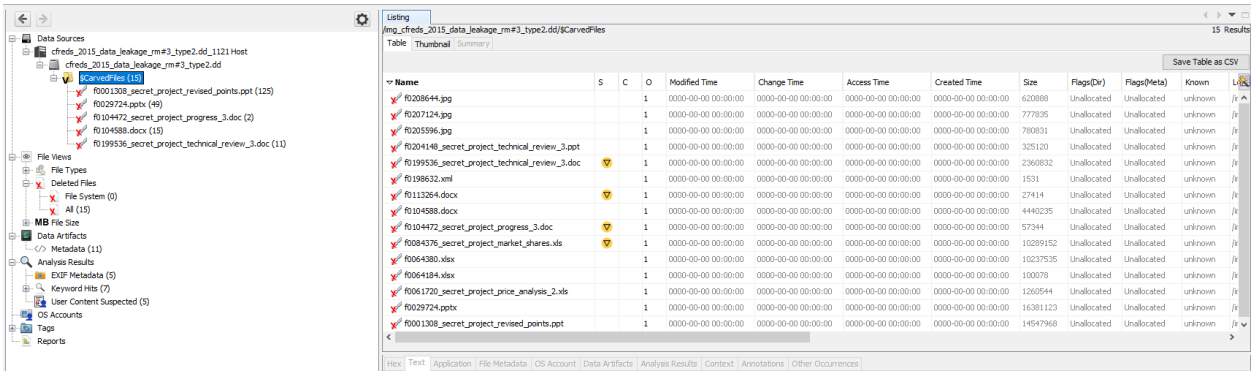
Hex | Text | Application | File Metadata | OS Account | Data Artifacts | Analysis Results | Context | Annotations | Other Occurrences

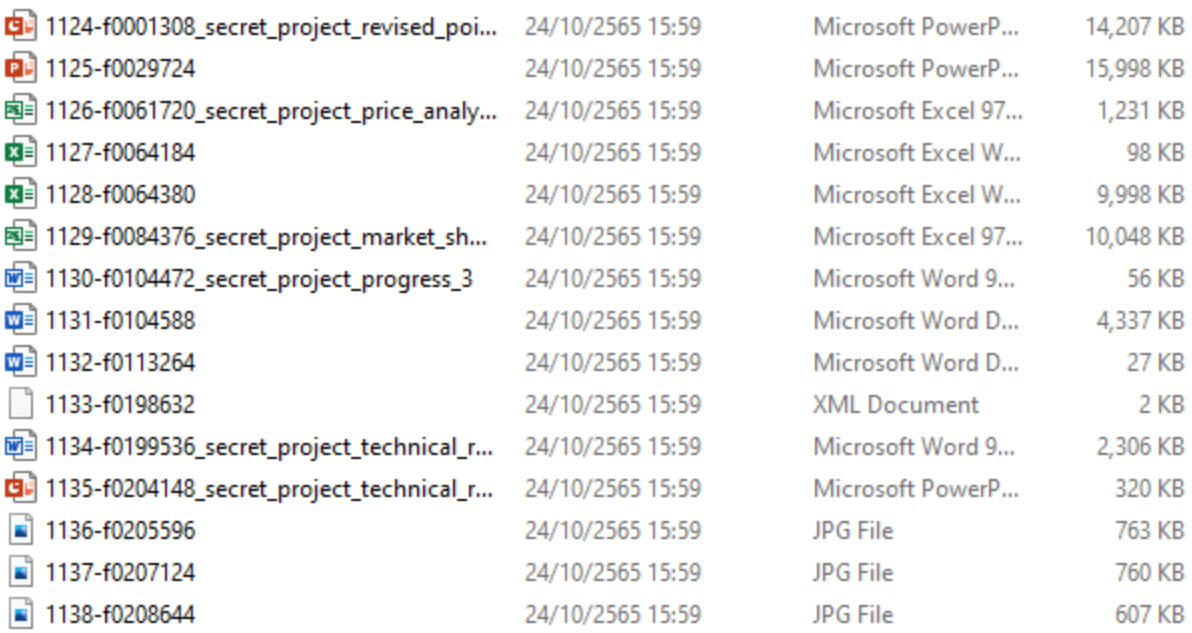
Result: 1 of 1 Result Metadata

Type	Value	Source(s)
Date Modified	2558-01-23 20:47:09 ICT	org.sleuthkit.autopsy.keyword
Program Name	Microsoft PowerPoint	org.sleuthkit.autopsy.keyword
Date Created	2547-05-06 01:10:13 ICT	org.sleuthkit.autopsy.keyword
User ID	company	org.sleuthkit.autopsy.keyword
Owner	company	org.sleuthkit.autopsy.keyword
Source File Path	/img_cfreds_2015_data_leakage_rm#2.dd/vol_vol2/$OrphanFiles/design/winter_storm.amr	
Artifact ID	-9223372036854775795	

- To prevent digital forensics, many files were renamed with wrong extension types before deleted. Nevertheless, file-types correction could solve the problems, and allowing them all to be readable.

5. Recover hidden files from the CD-R ‘RM#3’. What files were you able to recover?





6. What actions were performed for anti-forensics on CD-R ‘RM#3’?

Listing

/img_cfreds_2015_data_leakage_rm#3_type2.dd/$CarvedFiles

15 Results

Table | Thumbnail | Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Loca
f0001308_secret_project_revised_points.ppt			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	14547968	Unallocated	Unallocated	unknown	/img_
f0029724.pptx			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	16381123	Unallocated	Unallocated	unknown	/img_
f0061720_secret_project_price_analysis_2.xls			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	1260544	Unallocated	Unallocated	unknown	/img_
f0064184.xlsx			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	100078	Unallocated	Unallocated	unknown	/img_
f0064380.xlsx			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	10237535	Unallocated	Unallocated	unknown	/img_
f0084376_secret_project_market_shares.xls			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	10289152	Unallocated	Unallocated	unknown	/img_
f0104472_secret_project_progress_3.doc			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	57344	Unallocated	Unallocated	unknown	/img_
f0104588.docx			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	4440235	Unallocated	Unallocated	unknown	/img_
f0113264.docx			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	27414	Unallocated	Unallocated	unknown	/img_
f0198632.xml			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	1531	Unallocated	Unallocated	unknown	/img_
f0199536_secret_project_technical_review_3.doc			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	2360832	Unallocated	Unallocated	unknown	/img_
f0204148_secret_project_technical_review_3.ppt			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	325120	Unallocated	Unallocated	unknown	/img_
f0205596.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	780831	Unallocated	Unallocated	unknown	/img_
f0207124.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	777835	Unallocated	Unallocated	unknown	/img_
f0208644.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	620888	Unallocated	Unallocated	unknown	/img_

- Properties of all files associated with time were totally removed. But files’ contents were effortlessly accessible.